
title: "Savitribai Phule Pune University — Programming and Problem Solving"

code: "110005"

pattern: "2019"

total_marks: 70

duration: "2½ Hours"

Savitribai Phule Pune University

Programming and Problem Solving (110005)

2019 Pattern | Semester I/II

Total Marks: 70 | **Time:** 2½ Hours

Instructions: Q1/Q2, Q3/Q4, Q5/Q6, Q7/Q8. Neat diagrams must be drawn wherever necessary.

Unit III — Functions and Modules (18 Marks)

Q1) Attempt the following: [15]

- a) Define a **function**. Explain the need for functions in Python. [6]
- b) Differentiate between **local** and **global variables** with examples. [4]
- c) Write a program using a function to find the largest of 3 numbers by passing numbers as arguments. [5]

OR

Q2) Attempt the following: [15]

- a) Explain the following types of function arguments with examples: [6] i) **Default arguments**
ii) **Variable length arguments**
- b) Write a program to find the square of a number using **lambda function**. [5]
- c) What are **modules** and **packages** in Python? Explain a user-defined module with example. [4]

Unit IV — Strings (17 Marks)

Q3) Attempt the following: [15]

- a) Explain different ways of formatting a **string** in Python with suitable examples. [6]
- b) Explain different ways of **iterating a string**. [4]
- c) What is the output of the following for string `S = "Learning Python is fun"` : i) `print(S[:10])`
ii) `print(S[::-1])` iii) `print("Is" not in S)` iv) `print(S[4])` v)
`print(S[0:10])` [5]

OR

Q4) Attempt the following: [15]

- a) Explain the following string methods with examples: [6] i) `isspace()` ii) `join()`
iii) `lower()`
 - b) Write a program to display a string and count the number of characters in the string using a loop. [5]
 - c) Explain the `ord()` and `chr()` functions with examples. Explain `in` and `not in` operators on strings. [4]
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Unit V — Object Oriented Programming (18 Marks)

Q5) Attempt the following: [15]

- a) Explain the following **programming paradigms**: [6] i) **Monolithic Programming** ii) **Procedural Programming** iii) **Object Oriented Programming**
- b) Explain the following OOP concepts with examples: [5] i) **Class variable** ii) **Object variable**
- c) Write a Python program to create a class `Employee` with attributes `Name` , `emp_id` , and `salary` . Display data of 4 employees. [4]

OR

Q6) Attempt the following: [15]

- a) Explain the following features with examples: [6] i) **Containership** ii) **Inheritance**
iii) **Data Abstraction**
- b) Explain **class method** and **self object** with suitable example. [5]
- c) Write a program to create a class `Car` with two attributes `name` and `cost` . Create two objects and display their information. [4]

Unit VI — File Handling (17 Marks)

Q7) Attempt the following: [15]

- a) What is a **file**? Explain **relative** and **absolute path** of a file with examples. [6]
- b) Explain the following file handling methods: [5] i) `read(n)` ii) `readlines()`
iii) `close()`
- c) Explain the following file **access modes**: `r`, `w+`, `a+`, `rb`, `ab+`. [4]

OR

Q8) Attempt the following: [15]

- a) Explain any three **directory methods** with examples. [6]
- b) Explain **text** and **binary files** with examples. [5]
- c) Explain any three **dictionary methods** with examples. [4]

*** END OF QUESTION PAPER ***